

Part A: File System Navigation

Consider the file system below where the root is `~`, your home directory. The shapes following the names aren't part of the file system; they're pointers used in the questions below.

```
~
└─ spring ○
    └─ hist120A △
        │   └─ syllabus.txt ▲
        │   └─ essay1.txt
        │   └─ essay2.txt
        └─ math110
            │   └─ syllabus.docx ●
            │   └─ problem-sets
            │       └─ ps1.txt
            │       └─ ps2.txt
        └─ stat133
            └─ syllabus.docx
            └─ quizzes ▽
                └─ quiz1.qmd ■
                └─ quiz2.qmd
                └─ readme.txt
```

1. What is the absolute path to the file `●`?
2. What is the relative path from the directory `▽` to the file `■`?
3. What is the relative path from `▽` to `▲`?
4. If you are in a UNIX shell and your working directory is `○`, how can you print the content of `▲` using only one line of code?
5. If you are in a UNIX shell and your working directory is `△`, how can you create a new directory named `summer` in your home directory `~` using only one line of code?
6. If you are in a UNIX shell and your working directory is `▽`, how can you copy file `■` into your newly created `summer` folder using only one line of code?

Part B: File System in R (use the same tree above)

7. Starting from your home directory `~`, write R code to change the working directory to `Δ`.
8. After changing the working directory, write the R command to display your current working directory.
9. Assume your current working directory is `▽`. Write R code to read the file `▲` line by line into an object called `final`.

Part C: Working with Vectors and Matrices

10. Write an R command that creates the following vector (without using `c()`):

```
[1]  2  4  6  8 10 12 14 16
```

11. Predict the output of this code:

```
1:4 + 1:2
```

12. Predict the output of this code:

```
c(4,5,6) ^ c(1,2)
```

13. Predict the output of this code:

```
mean(c(T,T,FALSE,T,FALSE))
```

14. Using `z <- c("apple" = 2, "banana" = 4, "cherry" = 8)`, write the subsetting command that returns:

```
cherry banana apple
      8      4      2
```

(note: there are multiple ways to do this)

15. Using `z <- c("apple" = 2, "banana" = 4, "cherry" = 8)`, write the subsetting command that returns:

```
banana banana
      4      4
```

(note: there are multiple ways to do this)

16. What is the data type of the vector `TGIF <- c("movies", "chicken wings", 100L, NA, NULL)`?

17. Write the R command that creates a dimension `3x2` (rows by columns) matrix object named `mtx1`, using the vector `c(2,3,4,5,3,4)`, with the contents filled by column.

18. For the `mtx1` matrix you created in question 17, write the R command that reassign the element 5 to 2 (note: there are multiple ways to do this).

19. Predict the result (write it as a matrix) of the R command:

```
matrix(c(2,4,6,8),2,2)/2
```